

Experiment No. 7

(A) Maxwell Inductance Bridge Method

Aim: To determine the unknown inductance using Maxwell's inductance bridge method.

Apparatus:

1. NV6533 Trainer Board
2. 2 mm patch cords
3. Digital multimeter

Theory:

The Maxwell's inductance bridge is used for only determining medium inductance of given coil. Q factor cannot be measured by this method since we cannot bring the bridge to resonance condition. When the bridge is balanced, the equations for given circuit diagram are

$$Z_1 * Z_4 = Z_2 * Z_3$$

Where Z is impedance of each branch

$$[R_1 + j\omega L_1] * R_4 = [R_3 + j\omega L_3] * R_2$$

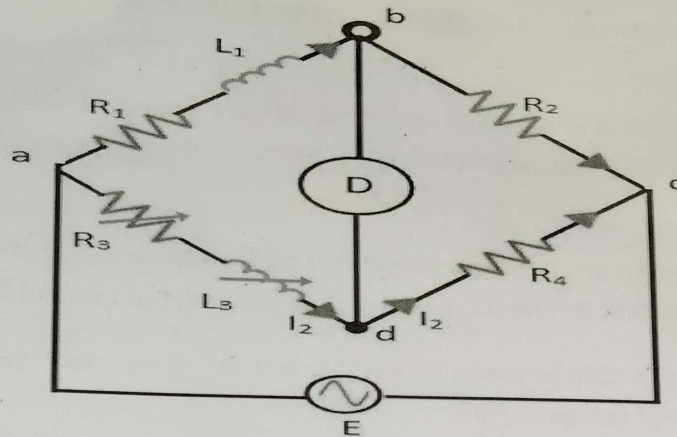
Equating real and imaginary parts, we get

$$R_1 = \frac{R_2 * R_3}{R_4}$$

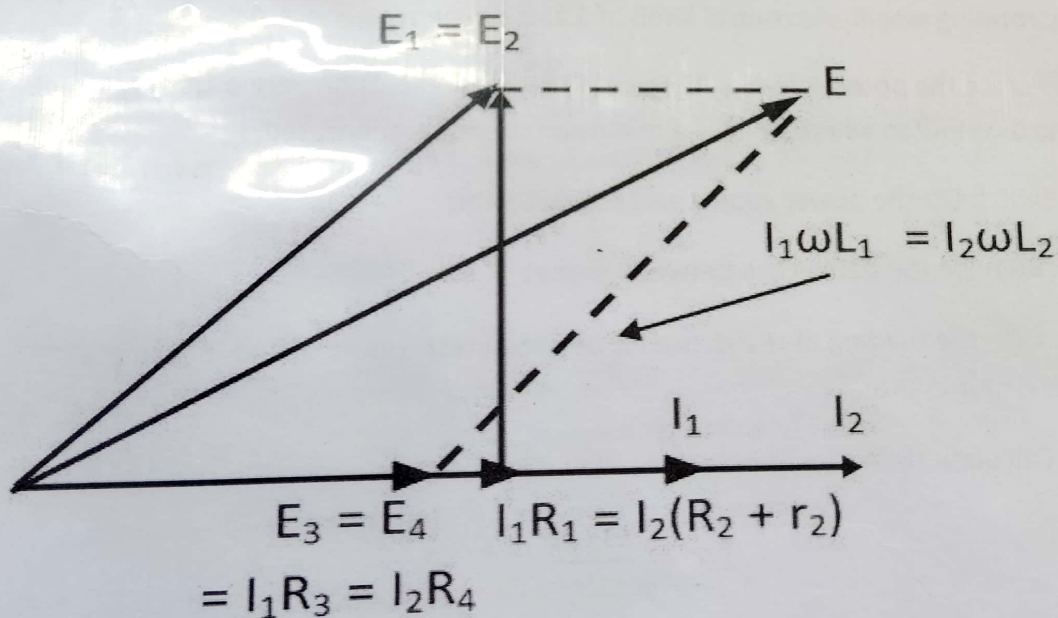
&

$$L_1 = \frac{L_3 * R_2}{R_4}$$

Circuit diagram:



Maxwell's Inductance Bridge



Phasor Diagram of Maxwell Inductance Bridge

Circuit Globe

Procedure:

1. Connect a patch cord between socket '13' of V_{in} terminal of Maxwell's inductance bridge and socket '29' of V_{out} terminal of the 1 KHz sine wave generator.
2. Connect a patch cord between socket '14' of V_{in} terminal of Maxwell's inductance bridge and socket '30' of V_{out} terminal of the 1 KHz sine wave generator.
3. Connect a patch cord between socket '1' and '2' and connect another patch cord between socket '8' and '11' to determine the value of L_{x1} and R_{x1} .
4. Connect a patch cord between socket '15' and '17' and socket '16' and '18' for the purpose of null detection.
5. Set the potentiometer R_2 in counter clockwise direction.
6. Switch on the power supply and the null detector.
7. Set the amplitude or loudness of the audio detector as per your requirement by rotating amplitude control knob of 1 KHz sine wave generator.
8. Rotate the potentiometer R_2 toward clockwise direction very precisely to find a condition where null (or a minimum sound) is generated.
9. Switch Off the power supply and null detector.
10. Remove the patch cord between socket '1' and '2'.
11. Take the reading of resistance R_2 between test -point '5' and '6' using a digital multimeters.
12. Calculate the value of inductance L_{x1} and resistance R_x using the formula

$$L_x = \frac{L_1 \cdot R_2}{R_4}$$

Where, $L_x = L_{x1}$, $L_1 = 12\mu H$, $R_4 = 100\Omega$

13. Calculate the value of unknown resistance using the formula

$$R_x = \frac{R_2 * R_3}{R_4}$$

Where, $R_3=470\Omega$, $R_4=100\Omega$

14. Connect a patch cord between socket '1' and '3' and another patch cord between socket '10' and '8' to determine the value of L_{x2} and R_{x2} .

15. Repeat the above step from 5 to 9.

16. Remove the patch cord between socket '1' and '3'.

17. Take the reading of resistance R_2 between test-point '5' and '6' using a digital multimeter.

18. Calculate the value of inductance L_{x2} and resistance R_x using the formula

$$L_x = \frac{L_1 * R_2}{R_4}$$

Where, $L_x=L_{x2}$, $L_1=12\mu H$, $R_4=100\Omega$

19. Calculate the value of unknown resistance using the formula

$$R_x = \frac{R_2 * R_3}{R_4}$$

Where, $R_3=470\Omega$, $R_4=100\Omega$

20. Now Connect a patch cord between socket '1' and '4' and another patch cord between socket '9' and '8' to determine the value of L_{x2} and R_{x2} .

21. Repeat the above step from 5 to 9.

22. Remove the patch cord between socket '1' and '4'.

23. Take the reading of resistance R_2 between test -point '5' and '6' using a digital multimeter.

24. Calculate the value of inductance L_{x2} and resistance R_x using the formula

$$Lx = \frac{L1 * R2}{R4}$$

Where, $Lx=Lx_2$, $L1=12\mu H$, $R4=100\Omega$

25. Calculate the value of unknown resistance using the formula

$$Rx = \frac{R2 * R3}{R4}$$

Where, $R3=470\Omega$, $R4=100\Omega$

Observation table:-

S No.	$R2(\Omega)$	$R4(\Omega)$	$R3(\Omega)$	$L1(\mu H)$	$Lx (\mu H)$	$Rx (\Omega)$
1.						
2.						
3.						

Calculation:

Measured value of $R2$ is _____ Ω

Now measured value of Lx by the formula

$$Lx = \frac{L1 * R2}{R4}$$

Measured value of resistance Rx by multimeter between sockets _____ Ω

Now measured value of Rx by the formula

$$Rx = \frac{R2 * R3}{R4}$$

$$R_{31} = 470 \Omega$$

$$L1 = 12$$

समस्या की कामकाज में आम-तौर पर प्रयोग में आने वाले वाक्यांश

Precautions:

1. Connect the circuit diagram with proper connection and take the measurements without error.
2. Calculate the values of inductor without any error

Result:

The unknown value of inductance L_{x1} = _____ μH

The unknown value of inductance L_{x2} = _____ μH

(B) Maxwell Inductance Capacitance Bridge Method

Aim: To determine the unknown Inductance and Q-factor using Maxwell's Inductance Capacitance bridge method

Apparatus:

1. NV6533 Trainer Board.
2. 2 mm patch cords.
3. Digital Multimeter.

Theory:

The Maxwell's inductance capacitance bridge is used for determining inductance and Q factor of given coil. As the variable capacitor is difficult to manufacture, thus this bridge is used to measure low Q factor (less than 10). When the bridge is balanced, the equations for given circuit diagram are

$$Z_1 * Z_4 = Z_2 * Z_3$$

Where Z is impedance of each branch

Equating real and imaginary parts, we get

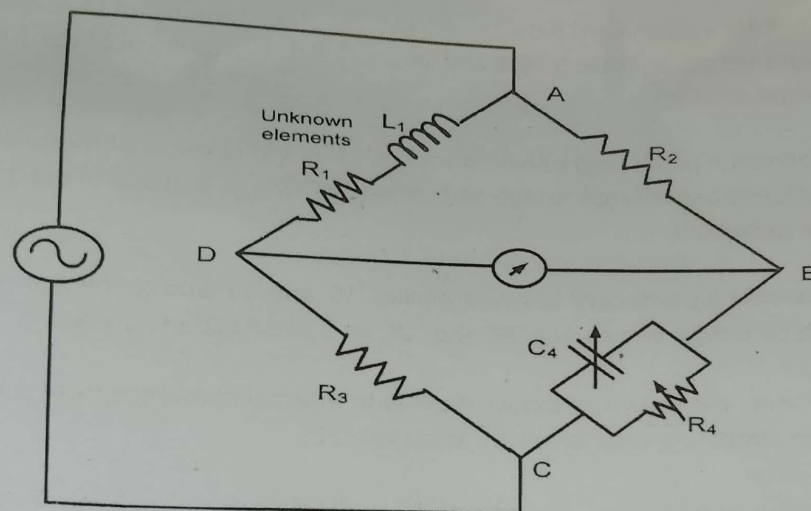
$$R_1 = \frac{R_2 * R_3}{R_4}$$

$$L_1 = R_2 * R_3 * C_4$$

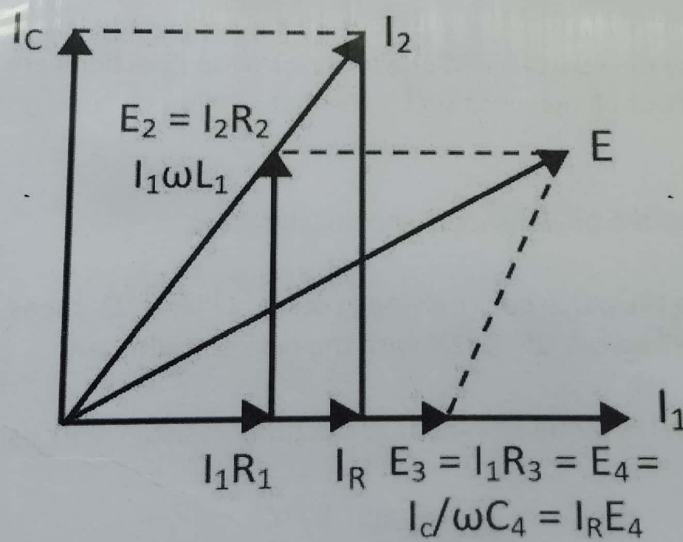
And Q factor is given by

$$Q = \frac{\omega * L_1}{R_1} = \omega * C_4 * R_4$$

Circuit Diagram :-



Maxwell inductance Capacitance Bridge



Phasor Diagram of Inductance
Capacitance Bridge

Circuit Globe

Procedure:

1. Connect a patch cord between socket '26' of V_{in} terminal of Maxwell's inductance capacitance bridge and socket '29' of V_{out} terminal of the 1 KHz sine wave generator.
2. Connect a patch cord between socket '27' of V_{in} terminal of Maxwell's inductance capacitance bridge and socket '30' of V_{out} terminal of the 1 KHz sine wave generator.
3. Connect a patch cord between socket '19' and '17' and connect another patch cord between socket '20' and '18' for null detection purpose.
4. Connect the unknown inductor L_{x4} with internal resistance from socket '22' to arm consisting resistances R_5 at socket '21'.
5. Set the amplitude or loudness of the audio detector as per your requirement by rotating amplitude control Knob of 1 KHz sine wave generator.
6. Set the potentiometer R_7 in full counter clockwise direction.
7. Switch on the power supply and the null detector.
8. Now vary the resistance R_7 towards clockwise direction very precisely with the help of pot till null position (or the first minimum sound position) is achieved.
9. Switch Off the power supply and null detector.
10. Remove the patch cord between socket '22' and '21' measure the resistance R_7 across '25' & '28' with the help of multimeter.
11. Calculate the value of inductance L_{x1} and resistance R_x using the formula

$$L_x = R_5 * R_7 * C_1$$

Where, $L_x = L_{x4}$, $R_5 = 221\Omega$, $C_1 = 330 \mu f$

12. Calculate the value of unknown internal resistance by using the following equation

$$R_x = \frac{R_5 * R_7}{R_6}$$

Where, $R_x = R_{x1}$, $R_5 = 221\Omega$, $R_6 = 1.122 K\Omega$

13. Calculate the value of Q-factor by using the following equation

$$Q = \frac{\omega * L_x}{R_x} = \omega * C_1 * R_6$$

Where, $\omega = 2\pi f$

14. Verify the result for the calculation of Q-factor using both the formula in the above step.

15. Connect the unknown inductor L_{x5} from socket '23' to arm consisting resistance R_5 at socket '21'.

16. Repeat the above step from 5 to 9.

17. Remove the patch cord between socket '23' and '21' measure the resistance R_7 across '25' and '28' with the help of multimeter.

18. Calculate the value of unknown inductance using the formula

$$L_x = R_5 * R_7 * C_1$$

Where, $L_x = L_{x4}$, $R_5 = 221\Omega$, $C_1 = 330\mu f$

19. Calculate the value of unknown internal resistance by using the following equation

$$R_x = \frac{R_5 * R_7}{R_6}$$

Where, $R_x = R_{x1}$, $R_5 = 221\Omega$, $R_6 = 1.122 K\Omega$

20. Calculate the value of Q-factor by using the following equation

$$Q = \frac{\omega * Lx}{Rx} = \omega * C1 * R6$$

Where, $\omega = 2\pi f$

21. Verify the result for the calculation of Q-factor using both the formula in the above step.

22. Connect the unknown inductor Lx5 from socket '23' to arm consisting resistance R5 at socket '21'.

23. Repeat the above step from 5 to 9.

24. Remove the patch cord between socket '24' and '21' measure the resistances R7 across 25 & 28 with the help of multimeter

25. Calculate the value of unknown inductance using the formula

$$Lx = R5 * R7 * C1$$

Where, $Lx = Lx4$, $R5 = 221\Omega$, $C1 = 330\mu F$

26. Calculate the value of unknown internal resistance by using the following equation

Where, $Rx = Rx1$, $R5 = 221\Omega$, $R6 = 1.122K\Omega$

27. Calculate the value of Q-factor by using the following equation

$$Q = \frac{\omega * Lx}{Rx} = \omega * C1 * R6$$

Where, $\omega = 2\pi f$

28. Verify the result for the calculation of Q-factor using both the formula in the above step.

Observation Table:

S.No	R7(Ω)	R5(Ω)	R6(Ω)	C1(μ F)	Lx(μ H)	Rx(Ω)	Q factor
1.							
2.							
3.							

Calculation:

Measured value of R7 is _____ Ω

Now measured value of Lx by the formula

$$Lx = R5 * R7 * C1$$

Measured value of resistance Rx by Multi-meter between sockets _____ Ω

Now measured value of Rx by the formula

$$Rx = \frac{R5 * R7}{R6}$$

Now measured value of Q factor by the formula

$$Q = \frac{\omega * Lx}{Rx} = \omega * C1 * R6$$

Precautions:

1. Connect the circuit diagram with proper connection and take the measurements without error.
2. Calculate the values of inductor and Q factor without any error

Result:

The unknown value of inductance L_{x4} , Resistance R_{x1} and Q factor are _____

The unknown value of inductance L_{x5} , Resistance R_{x2} and Q factor are _____

The unknown value of inductance L_{x6} , Resistance R_{x3} and Q factor are _____

Questions:

1. What are the Types of Maxwells Bridge?
2. List the Advantages of Maxwells Bridges?
3. List the Disadvantages of Maxwells Bridge?
4. what are the Applications of Maxwells Bridge?
5. Describe the detectors used for AC bridge?
6. What is the range of Q.?
7. What is meant by Q factors of the coil?